

$$\mathbf{s} \, = \, \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{a} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^9, \qquad \boldsymbol{\rho} \, = \, \begin{cases} 2 \, \mathrm{Im}(q_{t-1}^{-1} \otimes q_t) & \text{Madgwick on} \\ \mathbf{0} & \text{otherwise} \end{cases}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{10^6} \underbrace{\begin{bmatrix} \sigma_x & 0 \\ 0 & \sigma_y \end{bmatrix}}_{\text{sensitivity}} \underbrace{\begin{bmatrix} \mathbf{w}_x^\top \\ \mathbf{w}_y^\top \end{bmatrix}}_{2 \times 9 \text{ mix}} \mathbf{s}, \qquad \mathbf{w}_x, \mathbf{w}_y \in [-2000, 2000]^9$$

$$r \, = \, \sqrt{x^2 + y^2}, \qquad \begin{bmatrix} x' \\ y' \end{bmatrix} \, = \, \max\Big(0, \, \frac{r-d}{r}\Big) \, \begin{bmatrix} x \\ y \end{bmatrix}$$

$$g(v) \, = \, \mathrm{sgn}(v) \big(k_{\mathrm{lo}} \, |v| + |v|^{k_{\mathrm{exp}}} \big), \qquad k_{\mathrm{lo}} = 600, \, k_{\mathrm{exp}} = 1.2$$

$$e_t^{x,y} \, = \, \alpha \, g(x',y') \, + \, (1-\alpha) \, e_{t-1}^{x,y}, \qquad \alpha = 0.22$$

$$\begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \text{clamp}_{\pm 127} \Big(\lfloor e_t^{x,y} \rfloor \Big) \in \{-127, \dots, 127\}^2$$